Architecting Next-Gen Passive Drone Interceptor Systems

Exploiting Undetectable Signatures for Real-Time Tracking and Neutralization

Core Objective: Shifting from broadcast-reliant detection to passive physical signature exploitation

GNSS Shadows

Exploiting signal blockages and multi-path disturbances to map movement in real-time.

Internal Hardware Emissions

Detecting non-RF emanations and thermal or electromagnetic leakages from onboard circuitry.

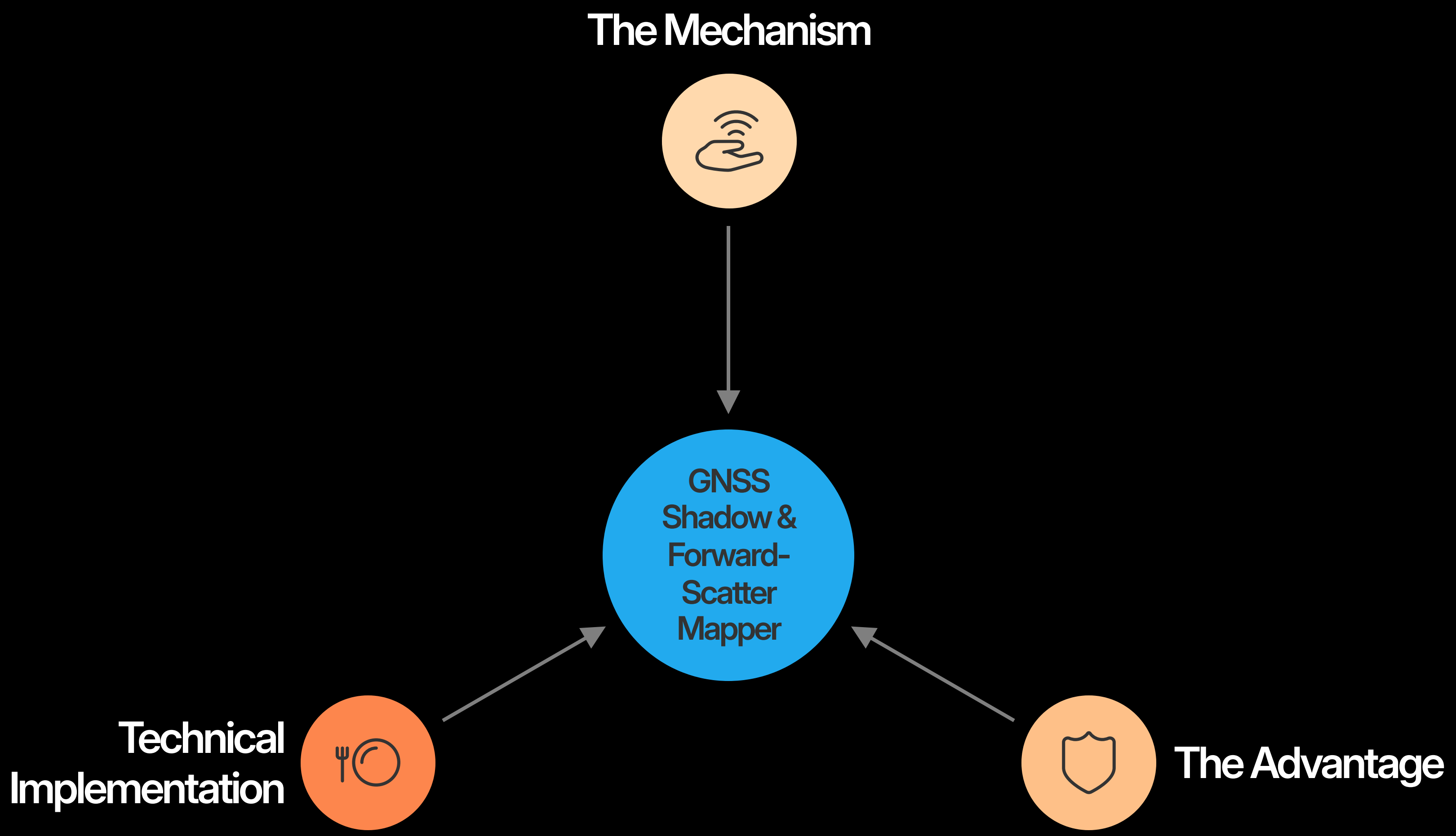
Acoustic Resonance

Utilizing advanced micro-array sensors to detect distinct propulsion frequency profiles.

PARADIGM SHIFT: MOVING BEYOND SIMPLE RF SNIFFING TO MULTI-MODAL SENSOR FUSION.

Signature 1: The GNSS Shadow & Forward-Scatter Mapper

Exploiting the GPS Constellation as a Passive Illuminator



Technical Overview

Signature 2: CMOS Clock ELINT Tracker

The Mechanism

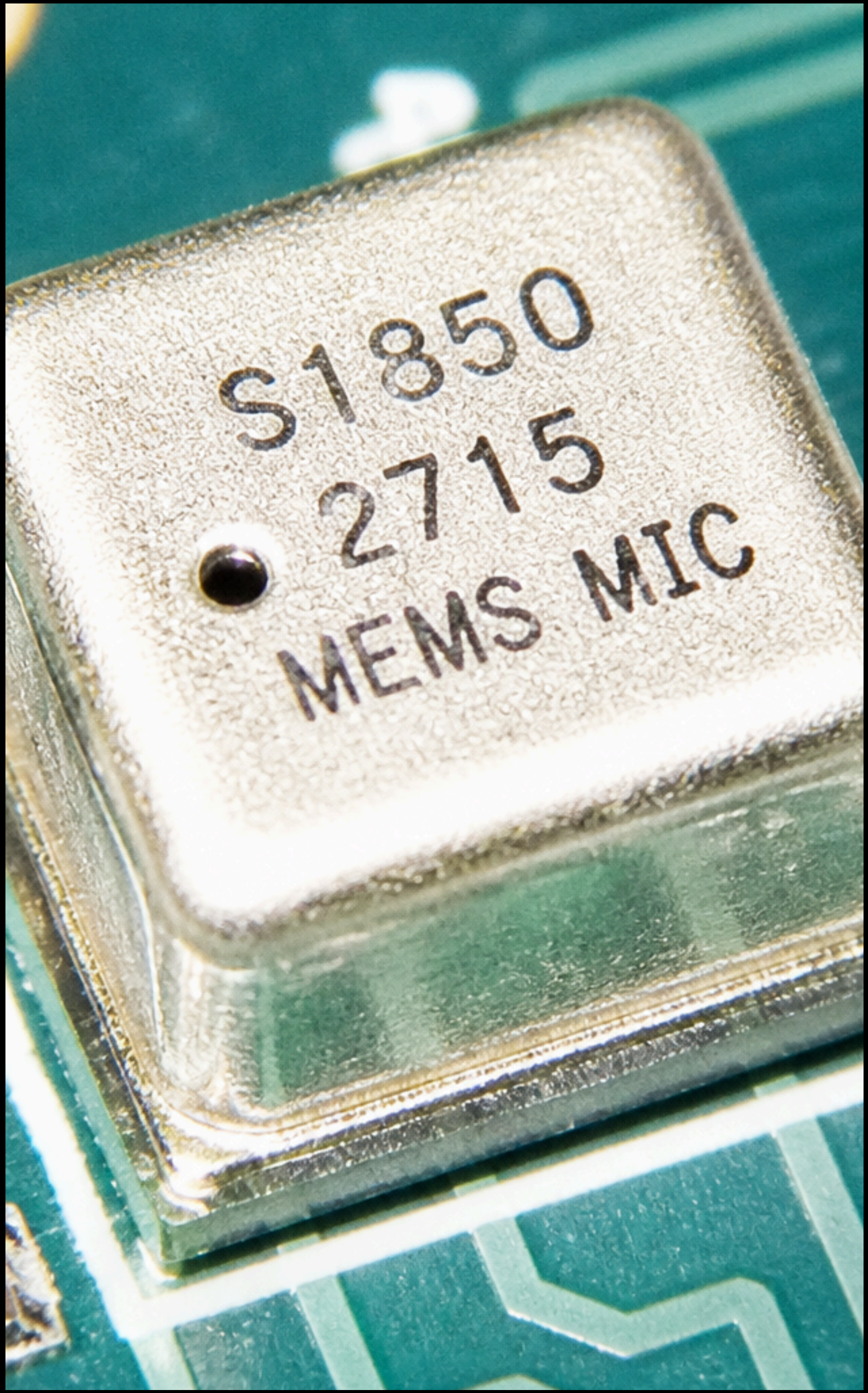
- Camera sensors emit constant pixel clock harmonics (24–72 MHz).

The Advantage

- A 'homing beacon' that persists even when video transmissions are disabled.

System Requirements

- 4-element crossed-dipole DF array + coherent SDR receiver.



Technical Specifications

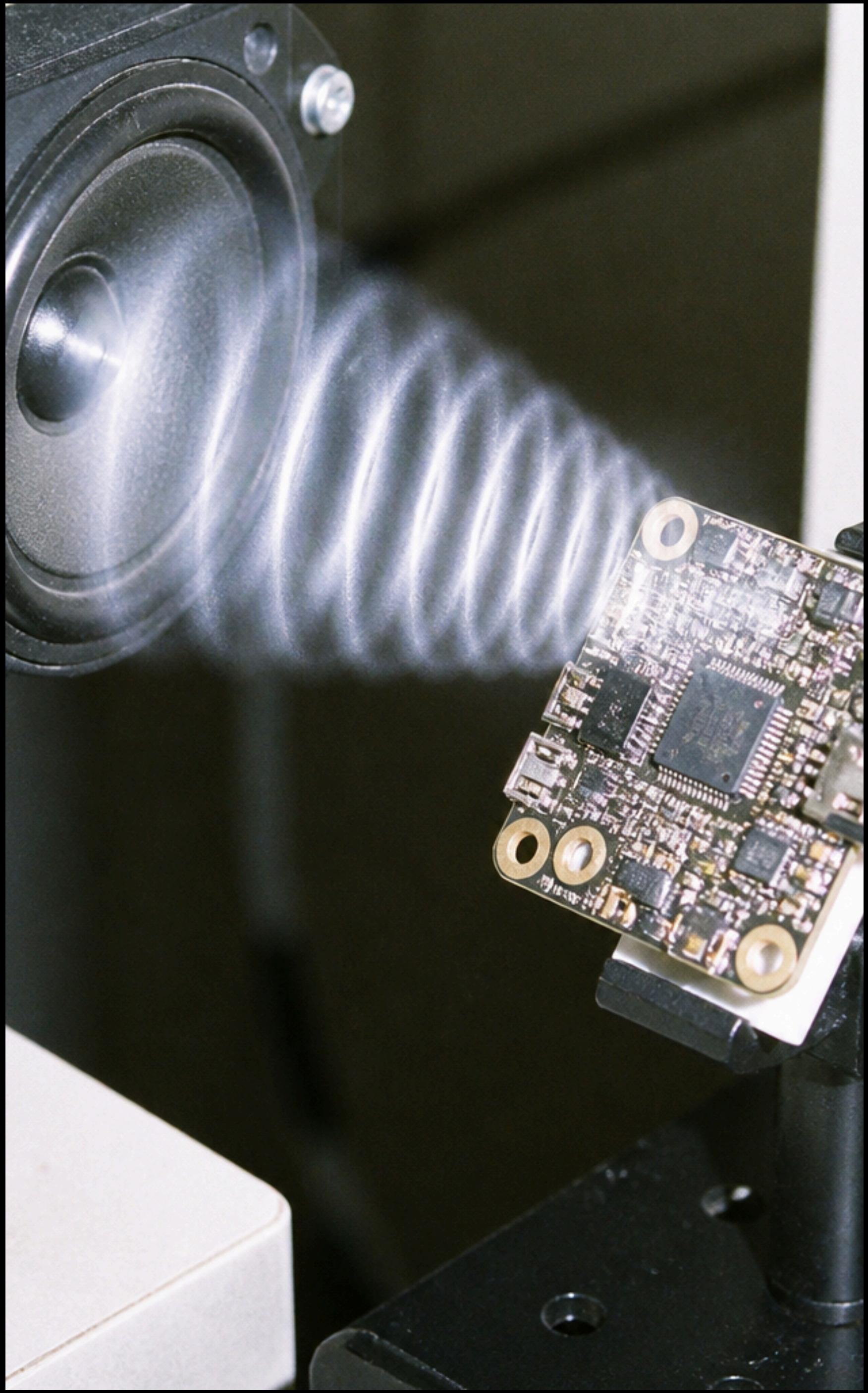
Signature 3: Ultrasonic Altimeter Passive Sonar

- **The Mechanism:** Most consumer drones emit 40kHz sonar pings.
- **The Advantage:** High sound pressure (100 dB @ 1m) makes for an easy 'shouting' signal.
- **Data Processing:** FPGA-based 16-channel beamforming to calculate 3D TDOA coordinates.

Data Matrix: Detection Modalities

Comparative Analysis of Unconventional Signatures

Signature	Detection Range	Primary Hardware	Passive
GNSS Shadow	3+ km	Coherent SDRs	✓
CMOS Clock	500m - 1km	VHF Array	✓
Ultrasonic Ping	150m - 250m	MEMS Mic Array	✓



Operational Overview

Unconventional Kill: The Ultrasonic Resonance Attack

"The sensor membrane is driven into saturation, forcing an immediate, irrecoverable tumble."

- **Mechanism:** Focus high-intensity (>140 dB) ultrasonic tones at the drone's MEMS gyro.
- **System Integration:** Use the same ultrasonic transducer array (repurposed from RX to TX).
- **Impact:** Immediate loss of attitude control with zero physical debris.

System Fusion: The Unified Passive Tracker

A Multi-Layer Tiered Engagement Workflow

1. Cueing

Ultrasonic arrays handle local zone coverage (<150m).

2. Identification

CMOS Clock DF confirms target drone type.

3. Precision Lock

GNSS Shadow mapping provides absolute ECEF coordinates.

4. Resolution

Optional autonomous kill engagement via phased-array acoustic burst.

Summary & Key Takeaways

Reinventing the Detection Perimeter

Go Passive

Emission-free sensors remain invisible to the target’s ESM systems.

Exploit Internal Noise

Every modern drone is built with 'hidden' unintentional radiators.

Hardware Focus

SDRs, FPGAs, and MEMS microphones represent the new frontline of signal intelligence.

Next Step: Start with a 24 GHz FMCW radar for base tracking, then layer in the passive sensors for high-fidelity confirmation.